PSC Project on Quantum Physical Unclonable Functions

A. Alice and Bob simulation

Objective for next time: Your goal for the next meeting is to simulate *Alice and Bob* (not Eve).

B. Choice of parameters

We consider N controlled spatial modes at the input and N detected modes at the output. The main parameters are:

- N: number of spatial modes (start from 100, try to see how it changes for larger N, the simulation will get slow for N too large);
- n: mean number of photons sent per pulse (check for some photons per channel use);
- $\eta \in (0, 1)$: overall efficiency of the setup (optical losses + detector efficiency); (play a bit with it, around 50% is reasonable)
- $k \in \{1, \ldots, N\}$: index of the target output mode on the detector;
- λ_{dc}: mean number of spurious (dark) clicks per pulse.

C. Computation of the database

First, generate the matrix H:

- start with a simple example where H is unitary;
- then move to the more realistic case where H is a random matrix.

For each $k \in \{1, \ldots, N\}$, compute the corresponding challenge C_k.

Example (for $k = 3$). Define the target output vector

$$y_k = (0,\, 0,\, 1,\, 0, \ldots, 0)^{\mathsf{T}},$$

i.e. a vector with a 1 at position k and 0 elsewhere. Then compute

$$C_k = \frac{H^\dagger y_k}{\|H^\dagger y_k\|},$$

where $H^\dagger$ is the Hermitian adjoint of H. Make sure that all vectors are correctly normalized in your implementation. At the end of this step, you should have a database with a matrix H and a set of N pairs of C_k and y_k (challenge and response).

D. Simulation of one run of the protocol

- Randomly choose a target index $k \in \{1, \ldots, N\}$.
- Propagate the corresponding challenge through the medium:

$$\tilde{y_k} = HC_k.$$

- For each output mode i (i.e. each camera pixel), sample the detected counts as

$$N_{\mathrm{det},i} \sim \mathrm{Poisson}(\mu_i), \qquad \mu_i = \eta_{\mathrm{tot}}\, n\, |\tilde{y}_i|^2 + \lambda_{\mathrm{dc}}.$$

- Arrange the $N_{\mathrm{det},i}$ values on a square grid to simulate a camera frame. For example, for $N = 100$ you can represent the output as a 10×10 image, it would be nice to have something like in Fig. 1.

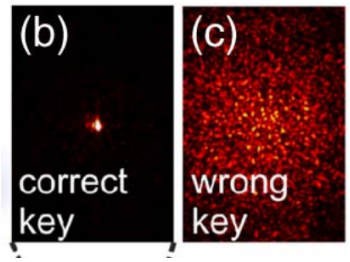

FIG. 1: screenshot from [1]

E. Estimation of the false rejection rate

By repeating the above simulation for many runs (with a legitimate user), you obtain a distribution of the detection counts and of the corresponding score. From this distribution, you can then estimate the *false rejection rate*.

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[1] Sebastianus A. Goorden, Marcel Horstmann, Allard P. Mosk, Boris Škorić, and Pepijn W. H. Pinkse. "Quantum-secure authentication of a physical unclonable key". Optica **1**, 421–424 (2014).